

Epidemiology and Healthcare Resource Utilization Associated With Children With Short Bowel Syndrome in the United States

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Abstract

Background: There is a lack of large database research relating to the epidemiology and health resource utilization associated with short bowel syndrome (SBS) in the United States. **Methods:** We analyzed the Kids' Inpatient Database for the year 2012 and utilized *International Classification of Diseases, Ninth Revision, and Clinical Modification (ICD-9-CM)* diagnosis codes to identify patients 0–3 years of age with SBS, who were matched by age and sex to children without SBS. The study variables included patient and hospital demographics, All Patient Refined Diagnosis Related Groups, in-hospital mortality, hospital length of stay, and hospitalization costs. We also determined the most frequent *ICD-9-CM* diagnostic and procedural codes associated with SBS. **Results:** Children with SBS demonstrated a higher rate of mortality than that of children without SBS (1.6% vs 0.7%; $P < .001$). Children with SBS also experienced a longer length of stay (median days [interquartile range]: 8 [15] vs 2 [3]; $P < .001$) and higher hospital costs (\$17,000 [\$34,000] vs \$3000 [\$5000]; $P < .001$). The most frequent medical diagnoses associated with SBS were infection (62%), anemia (29%), and liver disease (17%). Children with SBS also demonstrated more severe illness as assessed by an All Patient Refined Diagnosis Related Group score of 3 or 4 (94.30% vs 16.20%; $P < .001$). **Conclusions:** Children hospitalized with SBS have a high severity of illness and experience complicated inpatient courses related to their disease. Our study represents the first use of national U.S. data to study the epidemiology and health resource utilization associated with SBS. (*JPEN J Parenter Enteral Nutr.* 2017;41:878-883)

Keywords

short bowel syndrome; parenteral nutrition; Kids' Inpatient Database

Clinical Relevancy Statement

The epidemiology and health resource utilization associated with short bowel syndrome in children has not been studied with large database research. In this article, we analyze one of the largest pediatric hospital databases in the United States. Our results highlight the substantial burden of disease carried by children with short bowel syndrome and will aid in future research and healthcare planning relating to these patients.

Introduction

Short bowel syndrome (SBS) is a condition that usually results from the massive resection of the small intestine. In clinical practice, it implies the inability to maintain protein-energy, fluid, electrolyte, or micronutrient balances when on a conventionally accepted, normal diet, and it results in the need for parenteral supplementation to prevent nutrition or hydration deficits.¹ Congenital or acquired pediatric SBS is associated with significant morbidity and mortality rates.² A retrospective analysis by the Pediatric Intestinal Failure Consortium estimated that the cumulative incidences for enteral autonomy, death, and intestinal transplantation were 47%, 27%, and 26%, respectively.³

Unfortunately, few data derived from national registries are available regarding the epidemiology of SBS and health resource utilization related to this population in the United States. Previous studies were restricted to either single-center or relatively small multicenter collaborations.^{4,5}

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These data are thus unable to accurately portray the national disease burden and likely underrepresent the full disease impact carried by SBS in the pediatric population. Therefore, in this initial study, we interrogated a large pediatric hospital database with the aim of determining the extent to which healthcare resources are utilized by children with SBS who are 0–3 years of age.

Methods

Data Source

We obtained data for this study from the Kids' Inpatient Database (KID) of the Healthcare Cost and Utilization Project, sponsored by the Agency for Healthcare Research and Quality. The KID is a triennial database, and the 2012 iteration is the most current available version. It contains data from nonfederal, short-term, general, and specialty hospitals (including children's hospitals) accessible to the general public. These hospitals can be academic medical centers or other teaching hospitals. Individual hospitalizations—that is, discharge-level information (not patient level)—are de-identified and maintained in the KID as unique entries. Each discharge entry includes patient demographic details, 1 primary discharge diagnosis (based on the *International Classification of Diseases, Ninth Revision, and Clinical Modification* [ICD-9-CM] diagnosis codes), 1–24 ICD-9-CM secondary diagnoses, 1–15 ICD-9-CM procedural codes, hospitalization information, and hospital data. ICD-9-CM is a uniform and standardized coding system containing a multitude of codes: >14,000 diagnosis codes and 3900 procedure codes. ICD-9-CM diagnosis and procedure codes are an invaluable source of information for analyzing the cause for hospitalization and the procedures that patients undergo. The ICD-9-CM is maintained jointly by the National Center for Health Statistics and the Centers for Medicare & Medicaid Services. Individual-level population weights are used to generate national level estimates. All data in this study are reported as national-level estimates. For 2012, the KID contained 6,675,222 weighted total pediatric cases reported from 4179 hospitals in 44 states. Our final study cohort comprised 2680 children. The study excluded any protected health information and was deemed exempt from institutional board review.

Variable Definition

The primary variable of interest in this study was the presence of SBS, which does not have a unique ICD-9-CM code. Therefore, we used a combination of a single ICD-9-CM diagnostic code (579.3: other and unspecified postsurgical nonabsorption) and a single ICD-9-CM procedural code (99.15: parenteral infusion of concentrated nutrition substances). Any hospital discharge with the presence of both these ICD-9-CM codes was labeled in our study as an SBS case.

Study Demographics

Patient demographics, diagnoses, procedures, hospitalization data, and hospital information were obtained for the extracted cases. We also utilized All Patient Refined Diagnosis Related Groups (APR-DRGs), which classify patients by severity of illness (minor, moderate, major, or extreme). Patients with an APR-DRG score of 3 or 4 were deemed as severe disease.

Associated and Outcome Variables

The main outcome variables evaluated in this study were in-hospital mortality, hospital length of stay, and hospitalization costs (obtained by multiplying total charges by cost:charge ratios and rounded to the nearest \$1000). We also determined the most frequently associated diagnostic and procedural codes associated with SBS. Specifically, we extracted ICD-9-CM diagnoses codes that occurred at a frequency $\geq 5\%$ in children with SBS. Similarly, we extracted ICD-9-CM procedural codes that occurred at a frequency $\geq 3\%$ in patients with SBS.

Statistical Analysis

To compare the demographic characteristics and outcomes measures in children with and without SBS, we adjusted for the presence of confounders by using case-control matching involving high-dimensional propensity scores, which were generated by regression analysis of patients with SBS based on age and sex. Specifically, patients with SBS (cases) were matched by a high-dimensional propensity score based on a greedy matching algorithm to patients who did not have SBS (controls) with a 1:5 matching ratio. The control population did not have the presence of either ICD-9-CM code 579.3 or 99.15. This control population also excluded any discharge that was recorded under the category of an infant with a routine hospital birth. Statistical analyses were performed with SAS 9.2 (SAS Institute, Cary, NC). The Kolmogorov-Smirnov test was used to determine the normality of the data elements under study. The Kruskal-Wallis and χ^2 test were used to compare differences in continuous and categorical variables, respectively, between the 2 groups. The threshold for significance for all analyses was $P < .01$.

Results

Our study population consisted of children 0–3 years of age. We chose this age cutoff because it included 75.6% of the total number of SBS cases in the KID database. The incidence of SBS cases was observed to be much lower in children 4–18 years of age. Using data from the 2012 KID, we identified 2680 children with SBS (SE = 239). The median age of children with SBS was 1 year (interquartile range [IQR], 2 years), and 56.7% were male while 43.3% were female. This cohort of patients with SBS was then matched for age ($P = .9$) and

Table 1. Comparison of an Age-Matched and Sex-Matched Population of Hospitalized Children With and Without Short Bowel Syndrome.^a

Variable	Without Short Bowel Syndrome (n = 13,053)	With Short Bowel Syndrome (n = 2680)	P Value
Race			
White	49.10	39.90	<.001
Black	17.10	22.70	<.001
Hispanic	22.80	25.80	<.001
Other	11.00	11.70	.73
Insurance			
Private	38.50	19.80	<.001
Medicaid	54.70	70.90	<.001
None	6.60	9.30	<.001
Household income percentile			
\$1–\$37,999	32.40	34.30	.31
\$38,000–\$47,999	23.80	27.60	<.001
\$48,000–\$63,999	22.80	22.90	.38
\$64,000+	18.90	14.10	<.001
Hospital bed size			
Small	10.60	8.40	<.001
Medium	23.90	25.50	.07
Large	65.50	66.10	.6
Hospital location/characteristics			
Rural	8.40	0.30	<.001
Urban	91.60	99.70	<.001
Teaching hospital	70.10	95.90	<.001
Children's hospital	22.00	50.00	<.001
Disposition			
Home	92.60	61.90	<.001
Short-term hospital	2.80	5.10	<.001
Transfer to nursing facility	0.40	2.90	<.001
Home health	3.30	28.50	<.001
Patient characteristics			
Died	0.70	1.60	<.001
Severe illness	16.20	94.30	<.001
Chronic conditions	0 (1)	3 (3)	<.001
Diagnoses on record	4 (4)	9 (7)	<.001
Procedures on record	0 (1)	2 (4)	<.001
Length of stay, d	2 (3)	8 (15)	<.001
Hospital costs, \$	3000 (5000)	17,000 (34,000)	<.001

^aValues are presented as percentages or median (interquartile range).

sex ($P = .1$) in a 1:5 ratio to patients without SBS. Table 1 describes the comparison regarding patient and hospital demographics and the outcomes of these 2 groups of patients.

Children with SBS demonstrated a greater severity of illness as assessed by an APR-DRG score of 3 or 4 (94.30% vs 16.20%; $P < .001$), a greater number of chronic conditions (median [IQR], 3 [3] vs 0 [1]; $P < .001$), medical diagnoses (9 [7] vs 4 [4]; $P < .001$), and hospitalizations associated with a greater number of procedures (2 [4] vs 0 [1]; $P < .001$). There was a higher rate of mortality observed in children with SBS than in children without SBS (1.6% vs 0.7%; $P < .001$).

Children with SBS also experienced a longer length of stay (days, 8 [15] vs 2 [3]; $P < .001$) and higher hospital costs (\$17,000 [\$34,000] vs \$3000 [\$5000]; $P < .001$).

Children with SBS were less likely to have private insurance (19.80% vs 38.50%; $P < .001$) and more likely to have Medicaid (70.90% vs 54.70%; $P < .001$). There was no significant difference in the percentage of patients presenting to medium- ($P = .07$) and large-size hospitals ($P = .6$). However, children with SBS more frequently received their care in urban (99.70% vs 91.60%; $P < .001$), teaching (95.90% vs 70.10%; $P < .001$), and children's hospitals (50.0% vs 22.0%; $P < .001$).

Table 2. Most Common ICD-9-CM Diagnosis Codes Associated With Short Bowel Syndrome.

Code	Description	n	%
99932	Bloodstream infection secondary to central venous catheter	608	22.69
2859	Anemia unspecified	453	16.90
5738	Other specified disorder of the liver	444	16.57
7833	Feeding difficulties and mismanagement	443	16.53
7907	Bacteremia	395	14.74
99591	Sepsis	355	13.25
99931	Infection due to central venous catheter	310	11.57
53081	Esophageal reflux	296	11.04
9961	Mechanical complication of vascular device	295	11.01
78060	Fever, unspecified	288	10.75
78341	Failure to thrive	276	10.30
6910	Diaper or napkin rash	275	10.26
27651	Dehydration	271	10.11
2768	Hypopotassemia	237	8.84
78791	Diarrhea	215	8.02
2762	Acidosis	214	7.99
2761	Hyposmolality and/or hyponatremia	210	7.84
2809	Iron deficiency anemia, unspecified	200	7.46
2875	Thrombocytopenia, unspecified	188	7.01
7707	Chronic respiratory disease	154	5.75
78703	Vomiting alone	154	5.75
4659	Acute upper respiratory infection unspecified	144	5.37
99592	Severe sepsis	143	5.34
28529	Anemia of other chronic disease	137	5.11

ICD-9-CM, *International Classification of Diseases, Ninth Revision, and Clinical Modification*.

as compared with patients without SBS. Also, patients with SBS were less likely to receive a routine home discharge (61.90% vs 92.60%; $P < .001$) and more frequently received home healthcare at the time of discharge (28.50% vs 3.30%; $P < .001$).

Compared with the control population, white children composed a significantly lower proportion of SBS discharges (39.90% vs 49.10%, $P < .001$; odds ratio, 0.69; 95% confidence interval, 0.63–0.75), whereas black children had a disproportionately higher representation (22.70% vs 17.10%, $P < .001$; odds ratio, 1.4; 95% confidence interval, 1.3–1.6).

Tables 2 and 3 detail the most common ICD-9-CM diagnosis and procedural codes concurrently associated with a hospital discharge relating to SBS.

Discussion

This study uniquely investigated the epidemiology and health-care burden of young children hospitalized with SBS at a national level in the United States. A major strength of this study is the novel use of a nationwide database, which minimizes the effects of regional and population biases to which previous single-center and small multicenter studies were subject.^{4,5} To compare and contrast the demographics and

outcomes of patients with SBS, we selected an age-matched and sex-matched group of hospitalized children without SBS.

In these new data, we see that patients with SBS were overall much sicker than their counterparts without SBS. This was evidenced by a greater severity of illness based on APR-DRG scores, a greater number of medical comorbidities and diagnoses, and a more frequent undertaking of medical procedures in this cohort of patients. This fact likely explains the overall worse outcomes in children with SBS in terms of increased mortality, longer length of stay, and higher hospital costs. The effect of SBS on mortality is most likely a consequence of its predisposition to conditions such as sepsis and electrolyte abnormalities rather than a direct cause of death.

Black children had a disproportionately greater representation in the SBS cohort. It may be possible that this is linked to disparity in income; low socioeconomic status is a risk factor for preterm birth.⁶ In turn, prematurity and low birth weight are risk factors for SBS.⁷ A substantial proportion of children with SBS had Medicaid coverage. There are several potential contributors to this, including lower socioeconomic status associated with preterm birth, coupled with the significant time burden of this disease, rendering parents in greater need of financial support and fewer opportunities to maintain full-time employment.⁷ Almost all of the children hospitalized with SBS received their care in teaching institutes, and half of these were

Table 3. Most Common ICD-9-CM Procedure Codes Associated With Short Bowel Syndrome.

Code	Description	n	%
3893	Venous catheterization	633	23.62
9904	Transfusion of packed cells	493	18.40
3897	Central venous catheter placement	472	17.61
966	Enteral nutrition	465	17.35
4562	Partial resection of small intestine	324	12.09
8605	Foreign body or device removal from skin and subcutaneous tissue	264	9.85
5459	Lysis of peritoneal adhesions	247	9.22
9672	Continuous invasive mechanical ventilation ≥ 96 h	208	7.76
9604	Insertion of endotracheal tube	191	7.13
4651	Closure of stoma of small intestine	165	6.16
9702	Replacement of gastrostomy tube	134	5.00
4319	Other gastrostomy	133	4.96
9671	Continuous invasive mechanical ventilation ≤ 96 h	117	4.37
8607	Insertion of totally implantable vascular access device	113	4.22
3891	Arterial catheterization	100	3.73
4679	Other repair of intestine	95	3.54
9921	Injection of antibiotic	92	3.43
9749	Removal of other device from thorax	90	3.36
4620	Ileostomy, not otherwise specified	85	3.17
9905	Transfusion of platelets	83	3.10

ICD-9-CM, International Classification of Diseases, Ninth Revision, and Clinical Modification.

at children's hospitals. This is reflective of a referral bias and specialists concentrations that deal with this complex problem and so highlights the complexity of care required for these patients. While the majority of children without SBS received a routine discharge to home, children with SBS were much less likely to do so and instead be discharged with home healthcare or to other short-term and long-term care facilities.

Indeed, when we queried for medical conditions closely associated with SBS, catheter-related infections were observed for more than one-third of all patients with SBS. If the codes for bacteremia and sepsis were included, this number increased to almost two-thirds of all hospitalizations with SBS. Central line-associated bloodstream infections are well recognized in the setting of SBS and may lead to several complications, including growth failure, cholestasis, liver disease, and even death.⁸⁻¹⁰

Anemia from all causes was noted in approximately 30% of SBS cases. The presence of anemia in the setting of SBS has several potential etiologies—including nutrition deficiencies such as iron and vitamin B12, chronic blood loss from anastomotic site ulcerations in the small bowel, and mechanical trauma from indwelling central lines or in association with chronic disease. The presence of liver disease was also frequently associated with SBS in our cohort of patients. Intestinal failure-associated liver disease frequently occurs in children receiving long-term parenteral nutrition. In its progressive state, it may lead to cirrhosis, portal hypertension, and liver failure requiring liver transplantation. The pathogenesis is multifactorial and may involve prematurity, low birth weight,

duration of parenteral nutrition, SBS requiring multiple abdominal surgeries, and repeated episodes of sepsis. Deficiencies of taurine, cysteine, or choline may exacerbate intestinal failure-associated liver disease.¹¹

Medical procedures undergone by children with SBS in this study closely mirrored the associated complications above. For example, central line access and transfusion of packed red blood cells were the 2 leading procedures in this patient cohort. Surgical procedures involving the small intestine were noted with a cumulative frequency of approximately 35%. Finally, 12% of hospitalizations with SBS required mechanical ventilation for some duration of time.

This study has several limitations. We relied solely on ICD-9-CM codes for case identification. In the absence of specific ICD-9-CM codes for SBS, we used surrogate codes, which may have included other surgical conditions of the gastrointestinal tract that required parenteral nutrition. We could not correlate cases with clinical confirmation of disease given the lack of this information in the database. The KID does not track individual patients; therefore, it was not possible to determine recurrences or readmissions related to SBS. Finally, the costs associated with SBS reported in this study are likely to vary hospital and geographic region and hence should be interpreted with caution.

Conclusions

Children hospitalized with SBS have a complex disease with a high severity of illness and are especially vulnerable to poor

outcomes. Central line infections, anemia, and liver disease are frequently associated complications of SBS. Patients with SBS also frequently undergo surgical procedures related to the small intestine and often require intensive care unit admissions. Overall, this group of patients utilize a substantial proportion of healthcare resources in the United States. It is hoped that improvements in parenteral nutrition—specifically, the judicious use of lipid formulations—and the adoption of lock therapy for central lines will help decrease the incidence of SBS complications and, thus, the need for hospitalization. Our study represents a unique use of national U.S. data to study SBS in children and should aid in the planning of future research in this field.

Statement of Authorship

C. Pant, T. J. Sferra, and R. Gilroy equally contributed to the conception and design of the research; R. T. Fisher and M. Olyae contributed to the design of the research; C. Pant, T. J. Sferra, and R. Gilroy contributed to the acquisition of the data; C. Pant contributed to interpretation of the data. All authors contributed to the analysis of the data, drafted the manuscript, critically revised the manuscript, agree to be fully accountable for ensuring the integrity and accuracy of the work, and read and approved the final manuscript.

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